

Derivation and Numerical Reproduction of Liang, Goldsman, Mayergoyz & Oldiges (1997): 2-D Deterministic SHE-Boltzmann MOSFET Simulation

*Complete report, Revision 6 — Phase 1 (derivation) through the controlled
spatial-convergence study of the impact-ionization rate*

Reproduction study (code commit 676e74f+; Python 3.13.15, NumPy 2.5.1, SciPy 1.18.0)

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Abstract

This report derives and numerically reproduces the deterministic spherical-harmonic (SHE) Boltzmann device solver of Liang *et al.*, *IEEE Trans. Electron Devices* **44**(2), 257–267 (1997). Revision 2 superseded the earlier one-way/post-processed description: impact ionization is now *inside* the assembled electron collision operator (primary depletion + secondary redistribution), the self-consistent SHE $\leftrightarrow$ Poisson $\leftrightarrow$ hole-continuity loop is *demonstrated and observably converging*, global number- and terminal-current-conservation diagnostics are in place, and a physics-split preconditioner removes the previous solver bottleneck (~ 10 min $\rightarrow \sim 15$ s on the 40×34 problem). Established results: core transport (n , $|v|$) is validated and spatially robust (T_e moderately grid-sensitive); the self-consistent feedback materially suppresses avalanche generation and drain current; conservation holds across one-way, refined, and coupled runs. Open items are stated precisely: the impact-ionization magnitude is *not spatially converged* — a 2×2 grid $\times$ coupling study drives $G_{ii,\max}$ multiplicatively from 1.35×10^{27} to 5.24×10^{26} cm $^{-3}$ s $^{-1}$ (the finest-tested coupled value, not a continuum estimate), while a one-way energy refinement to $\Delta H = 6.25$ meV moves it only -2.4% (negligible beside the spatial error); the T_e/G_{ii} peak offset is unresolved; and the device has been recalibrated to $V_{th} = 0.614$ V for the quantitative Liang overlays. The robust reproductions are the transport moments (n , $|v|$). **Revision 5 replaces the damped-Picard coupling with Anderson acceleration on a frozen H -grid, which converges the self-consistent fixed point that Picard could not** (§10, Fig. 4): on each grid the true residual $\max |G(\phi) - \phi|$ falls to a ~ 3 –4 mV inner-solve floor and G_{ii} settles to a residual-floor last-iterate value (last-4 spread 1.6–11%; not yet the Rev. 6 observable plateau), where the 50-iteration Picard baseline left G_{ii} in monotone free-fall. This separates two distinct questions: the *nonlinear-loop* convergence is now resolved, but the converged fixed-point G_{ii} is *still spatially unconverged* — over the $40\times 34 \rightarrow 60\times 50 \rightarrow 80\times 66$ sequence it decreases $8.1 \rightarrow 2.2 \rightarrow 1.5\times 10^{26}$ cm $^{-3}$ s $^{-1}$ (decelerating). Crucially, this converged $G_{ii} \approx 1.5\times 10^{26}$ is only 7% of Liang’s top contour — an order of magnitude below the paper — so the earlier snapshot “matches” (27%, 58%) were artefacts of an under-converged loop, and the honest converged result is that our surrogate Keldysh II rate on the reconstructed device produces far *less* avalanche than Liang’s numerical rates. The overlays (§11) use the converged finest-grid (80×66) fields: $T_{e,\max} \approx 3201$ K (64% of Liang’s top 5000 K contour) and $G_{ii,\max} \approx 1.5\times 10^{26}$ (7%), both drain-localized with T_e far broader than G_{ii} ; a one-way *SHE velocity-moment* I-V reproduces Liang’s Fig. 8 method (terminal current from the SHE face flux, not a mobility model, S/D conservation $\leq 10^{-3}$, running 0.5 – $0.96\times$ the DD proxy); and a per-plot comparison table is given for each of Liang’s Figs. 2–8. **Revision 6 carries out the controlled spatial-convergence study that Rev. 5 left open** (§12). Two findings are firm. First, the earlier uniform sequence was itself *under-resolving* the drain avalanche: a drain-graded 100×70 mesh drops $G_{ii,\max}$ about an order

of magnitude below the “finest uniform” 80×66 , and the spatially-integrated G_{tot} falls in step, so this is a genuine avalanche-magnitude effect, not peak-cell placement. Second — and decisively — a properly *controlled self-similar trio* $71 \times 58 \rightarrow 100 \times 70 \rightarrow 141 \times 85$ (fixed grading law, byte-identical frozen H -grid verified by SHA1, $\sqrt{2}$ interval refinement, all three fixed points converged to $O(10^{-4})$ plateau quality) is *significantly non-monotone*: $G_{ii,\text{max}} = 2.990 \rightarrow 2.135 \rightarrow 2.413 \times 10^{25} \text{ cm}^{-3} \text{ s}^{-1}$, with the down-then-up reversal exceeding the plateau bands by a heuristic ratio $S_{\text{rev}} \approx 874$ (826 for G_{tot}). A sign-changing trio does not support the leading-order single-power Richardson model, so extrapolation is suppressed: *spatial convergence is not demonstrated, and no continuum G_{ii} is inferred from this trio* (this does not prove a continuum limit fails to exist). A separate methodological result: a fixed outer-iteration budget does *not* guarantee equal fixed-point quality across meshes — the coarser nodes’ 60-iteration values were biased transients that shifted by up to +18% on continuation, so node validity is gated on an observable plateau, not an iteration count. The order- 10^2 gap to Liang’s top contour therefore stands as a not-converged tested-grid quantity, consistent with a substantial model/physics discrepancy but not yet a bounded continuum/model gap. **Revision 4** reframed the work honestly (retained here): it is a *reduced* P1/ $(\ell \leq 1)$ SHE-Boltzmann reproduction of the principal 2-D transport and impact-ionization *signatures*, using reconstructed device data and *surrogate scattering and band models*. Liang’s explicit surface-roughness, surface-acoustic, and ionized-impurity operators are *lumped* into a single calibrated momentum time τ_1 , not reproduced as distinct mechanisms (Table 2); the quantitative overlays use a *calibrated* device (V_{th} fitted to 0.614 V), so they are calibrated comparisons, not blind predictions (Table 3); and the DD I-V curve is retained only as a comparison proxy, while the one-way SHE face-flux sweep reproduces Liang’s current-extraction method (§11). Every model element is classified as *paper-specified, reconstruction, surrogate/lumped, or unconverged/mesh-dependent*.

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1 Overview and audit provenance

Liang *et al.* solve, self-consistently, three coupled equations on a 2-D MOSFET cross-section: Poisson, the electron BTE (first-order SHE), and hole continuity with an impact-ionization pair-generation source. This reproduction has passed through four external audit rounds and three code patches (Appendix B); the present revision documents the resulting coupled state and its quantitative comparison against Liang’s figures.

Honesty note. Model substitutions carried throughout and classified in Table 1: the doping profile is a *reconstruction*; the band is a 1-parameter Kane *surrogate* (paper: 2-band Fiegna/Brunetti); the impact-ionization rate is a Keldysh *surrogate*; the momentum time τ_1 is a single calibrated surrogate (not a summed Table-I model); and the device V_{th} , initially an over-high 1.64 V, is recalibrated to 0.614 V (§9) for the quantitative overlays (§11).

Part I

Method derivation and validated foundation

2 First-order SHE equation

With $\mathbf{E} = -\nabla_{\mathbf{r}}\phi$, expand $f = f_0(\mathbf{r}, \varepsilon) + \hat{\boldsymbol{\Omega}} \cdot \mathbf{f}_1(\mathbf{r}, \varepsilon)$ (all $\ell \geq 2$ discarded). This $\ell \leq 1$ (P1) truncation *is* Liang’s “first four spherical harmonics”: Y_0^0 plus the three $\ell = 1$ components ($Y_1^{-1,0,1}$) carried by $\hat{\boldsymbol{\Omega}} \cdot \mathbf{f}_1$. Projection gives the current relation and the conservative augmented-space balance

$$\mathbf{f}_1 = -\tau_1 v (\nabla_{\mathbf{r}} f_0 - q \mathbf{E} \partial_{\varepsilon} f_0), \quad \nabla_{\mathbf{r}} \cdot \mathbf{j}_x + \partial_{\varepsilon} j_{\varepsilon} = Z \mathcal{Q}[f_0] + Z S_{\text{gen}}, \quad (1)$$

$\mathbf{j}_x = -ZD(\nabla_{\mathbf{r}} f_0 - q \mathbf{E} \partial_{\varepsilon} f_0)$, $D = \frac{1}{3} v^2 \tau_1$. In total energy $H = \varepsilon - q\phi$ this becomes pure spatial diffusion at fixed H , the collision operator coupling $H \leftrightarrow H \pm E_{op}$ locally in space, over the global mesh $-q\phi_{max} \leq H \leq \varepsilon_{max} - q\phi_{min}$ with mask $0 \leq H + q\phi \leq \varepsilon_{max}$; $\varepsilon = 0$ reflecting, ε_{max} absorbing. The DOS-weighted acoustic operator $\mathcal{Q}_{ac} = Z^{-1} \partial_{\varepsilon} [ZA(\partial_{\varepsilon} + 1/k_B T)f_0]$ (Scharfetter–Gummel) makes the collision term irreducible; it is number-conserving (defect 7.6×10^{-17}) with the Maxwellian as its exact null.

3 Reconstructed device, taxonomy, foundation

The 0.35 μm LDD NMOS uses error-function junctions ($x_j = 0.170 \mu\text{m}$ metallurgical). Foundation checks (all at $T = 300\text{K}$): equilibrium Poisson (built-in $+0.586/-0.44\text{ V}$, quadratic Newton); nonparabolic $N_c = 2.857 \times 10^{19}$ (0.1% of the independent value; parabolic limit 0.09% of closed form); bulk scattering engine detailed-balance 2×10^{-5} , zero-field $T_e \rightarrow 309.7\text{ K}$, monotone heating. The DD solve at $V_{gs} = V_{ds} = 3\text{ V}$ gives the operating-point potential driving the SHE.

Table 1: Modelling taxonomy (Rev. 2 adds a *not-yet-converged* class).

Element	Class	Basis / status
Method eqs., SHE reduction, moments	paper	Liang <i>et al.</i>
Table-I params, spherical band	paper	paper Sec. II
$L_{\text{eff}}, t_{ox}, x_j, N_D^{\pm}$	paper	paper text
1-param Kane $\gamma(\varepsilon)$	surrogate	paper: 2-band Fiegna/Brunetti
LDD doping shape, mesh	reconstruction	ref. [25] (Khalil 1994) unavailable
Surface-roughness / -acoustic / impurity scattering	lumped surrogate	folded into τ_1 ; not distinct operators (Table 2)
Channel $N_A, \Phi_{gate}, \tau_1, c_{op}, 1/\tau_{ii}$	surrogate	calibrated; disclosed (Table 3)
$G_{ii, \text{max}}$ magnitude	loop-conv., mesh-dep.	Anderson fixed point 1.5×10^{26} (80×66); $\sim 15\times$ below Liang (spatial attribution superseded by §12)
T_e/G_{ii} peak offset	unconverged	unresolved at present mesh

Table 2: Paper-to-code physics matrix: every Liang scattering mechanism, and how it is treated here. “Lumped” means folded into the single calibrated momentum time τ_1 rather than implemented as a distinct spherical-harmonic collision operator.

Mechanism	Liang <i>et al.</i>	This work	Status
Acoustic phonon (bulk)	explicit elastic op.	DOS-weighted SG operator	present (reduced)
Optical / intervalley phonon	explicit inelastic op.	detailed-balance operator, surrogate c_{op}	present (surrogate)
Ionized impurity (Brooks–Herring)	explicit $\ell=1$ op.	folded into τ_1	lumped
Surface acoustic phonon (D_{sa})	explicit op.	folded into τ_1	lumped
Surface roughness (Γ, δ)	explicit op.	folded into τ_1	lumped
Impact ionization	numerical rate [23]	Keldysh surrogate, in operator	present (surrogate rate)
e – h recombination (SRH)	in BTE (trap occupancy)	SRH in DD hole continuity	proxy location
Net momentum relaxation	sum of the above	single calibrated τ_1	lumped surrogate

Table 3: Calibration vs. validation. Fitted parameters, the target they were fitted to, the fitting data, and an *independent* quantity used to check them. The quantitative overlays are therefore calibrated comparisons, not blind predictions.

Parameter	Fitted target	Fitting data	Independent quantity	comparison
$\Phi_{gate} = -0.74 \text{ V}$	$V_{th} = 0.614 \text{ V}$	extracted threshold	$I_d - V_g$	subthreshold ($\sim 99 \text{ mV/dec}$); swing overlay
$N_{A, \text{chimp}} = 4 \times 10^{17}$	V_{th}	threshold extraction	hot-spot location	surface inversion density
$\tau_1 = 10^{-14} / \sqrt{\gamma \gamma'}$	bulk mobility	bulk $ v - E$ curve	n_{surf}	device $ v _{\text{max}} \approx 10^7 \text{ cm/s}$
$c_{op} = 4.69 \times 10^{-34}$	$ v _{\text{sat}}$	bulk hot-electron T_e	device $T_{e, \text{max}}$	localization
$1/\tau_{ii} (P=2 \times 10^{13}, \varepsilon_{th}=1.1 \text{ eV})$	bulk $T_e(E)$ heating	Keldysh-type gate	G_{ii}	drain localization; feed-back fraction
	II onset threshold	surrogate		

Part II

The 2-D SHE core, impact ionization, and self-consistency

4 Assembled operator, impact ionization, and moments

The solver discretizes the $(\mathbf{r}, H)$ equation by box integration; contributions (spatial diffusion $\times \Delta H$, optical in/out $\times A_{ij} \Delta H$, acoustic SG flux $\times A_{ij}$) share units and form a positive-diagonal system. Contacts inject a Maxwellian $f_0 = n_{\text{contact}} e^{-\varepsilon/k_B T} / \int Z e^{-\varepsilon/k_B T} dH$.

Impact ionization in the electron BTE. An event removes a primary electron at rate $1/\tau_{ii}(\varepsilon)$ (Keldysh surrogate, $\varepsilon_{th} = 1.1 \text{ eV}$) and deposits two secondaries at $\varepsilon' = (\varepsilon - \varepsilon_{th})/2$, linearly distributed onto the neighbouring H nodes, with net electron gain +1 per event. This is *assembled into the electron collision operator* (primary depletion + secondary redistribution), so f_0 is directly shaped by avalanche; $G_{ii} = \int Z f_0 / \tau_{ii} d\varepsilon$ is a moment of that II-affected distribution. The equal kinetic-energy split of the two secondaries, and the fact that this scattering-in term leaves the f_0 shape essentially unchanged, follow Liang *et al.* (§IV) directly. The absorbing ε_{max} cutoff acts on spatial, acoustic, and optical fluxes. Moments n , $\langle \mathbf{v} \rangle$, T_e , G_{ii} , and the terminal currents are reconstructed from the *same finite-volume face fluxes* as the operator.

5 Numerics: physics-split preconditioner

The large-grid system is solved by a multiplicative physics-split preconditioner. Because DOFs are numbered with H fastest, same-spatial-cell couplings (acoustic + optical + II redistribution + spatial diagonal) form contiguous local energy blocks — the **E block** — factored cheaply; the diagonal plus cross-cell couplings form the fixed- H real-space **S block**. One application does $E^{-1} \rightarrow S^{-1} \rightarrow E^{-1}$ with a fresh residual between stages; the solved matrix is never shifted (shifts touch only preconditioner factors). On the 328 544-DOF 40×34 problem this solves to 10^{-10} in $\sim 15 \text{ s}$ (raw scaled residual 1.22×10^{-11}), where the previous global ILU did not finish its setup in 120 s. The legacy path is retained as `method="legacy_ilu"` for regression.

6 Numerical audit of the one-way baseline

Table 4 is the extracted audit of the locked one-way reference (commit `f016e3c`); it is preserved as a regression benchmark.

Table 4: Clean 40×34 numerical audit (impact ionization in the collision operator).

Quantity	Value	Location / note
$n_{\max}$	$1.0516 \times 10^{20} \text{ cm}^{-3}$	source n^+ (0.082, 0.043)
$T_{e,\max}$	3232.9 K	drain (0.595, 0.005)
$G_{ii,\max}$	$1.3513 \times 10^{27} \text{ cm}^{-3}\text{s}^{-1}$	drain (0.595, 0.009)
$ v _{\max}$	$9.732 \times 10^6 \text{ cm/s}$	drain surface (0.554, 0)
$T_e - G_{ii}$ peak separation	5 nm (~ 1 cell)	co-located within mesh
SHE currents I_{src}/I_{drn}	$-1.7044/ + 1.7071 \times 10^{-4} \text{ A}/\mu\text{m}$	mismatch 0.16%
global continuity residual	-5.70×10^{-6} (relative)	—
orphan states	0	no artificial sinks
f_0 pre-clip min	-3.3×10^{-11}	46/328544 tiny negatives

7 Grid refinement (40×34 vs 60×50 , one-way)

Refining the spatial mesh at fixed $\Delta H = 12.5 \text{ meV}$:

Quantity	40×34	60×50	converged?
$n_{\max}$	1.0516×10^{20}	1.0473×10^{20}	yes (-0.4%)
$ v _{\max}$	9.73×10^6	10.0×10^6	yes ($+2.7\%$)
$T_{e,\max}$ [K]	3232.9	3423.0	moderate ($+5.9\%$)
$G_{ii,\max}$	1.3513×10^{27}	8.26×10^{26}	no (-40%)
T_e/G_{ii} separation	1 cell	2 cells	grid-scale (unresolved)
current mismatch / residual	0.16% / 5.7×10^{-6}	0.07% / 7.6×10^{-6}	conserved

n and $|v|$ are spatially robust over this refinement; T_e is moderately grid-sensitive (5.9%, not converged); G_{ii} is *not* grid-converged (it depends on the high-energy tail); and the peak separation tracks the mesh, so it is unresolved, not a reproduced physical offset.

8 Self-consistent SHE $\leftrightarrow$ Poisson $\leftrightarrow$ hole loop

A damped outer iteration solves the SHE (with impact ionization) on the current potential, drives hole continuity with G_{ii} as pair generation (holding n_{SHE}), updates Poisson from the SHE density, damps, and repeats; n is recomputed from the SHE after each potential update (not constitutively from DD). Over 12 iterations (Fig. 2) the maximum potential update falls monotonically $17.7 \rightarrow 4.0 \text{ mV}$, and over these first 12 iterations the observables *appeared* to plateau: $G_{ii,\max}$ suppressed 37% ($1.35 \rightarrow 0.85 \times 10^{27}$), drain current 29% ($1.71 \rightarrow 1.21 \times 10^{-4} \text{ A}/\mu\text{m}$), $T_{e,\max} \sim 3240 \text{ K}$. **This early plateau is only apparent:** the extended 50-iteration study (§9, Fig. 3) shows G_{ii} continues to fall monotonically. This is the physically expected negative feedback: the SHE electrons and the G_{ii} -generated holes reshape the potential, lowering the peak drain field and throttling avalanche. Final conservation is clean (relative number balance 2.4×10^{-5} , currents balance 0.14%).

Honesty note. This is a 12-iteration *snapshot*. The loop is not converged ($\max |\Delta\phi| \approx 4 \text{ mV}$, above the 2 mV criterion), and the apparent G_{ii} plateau does *not* survive: extending to 50 iterations (§9) shows G_{ii} keeps decreasing. n and $|v|$ are the robust observables.

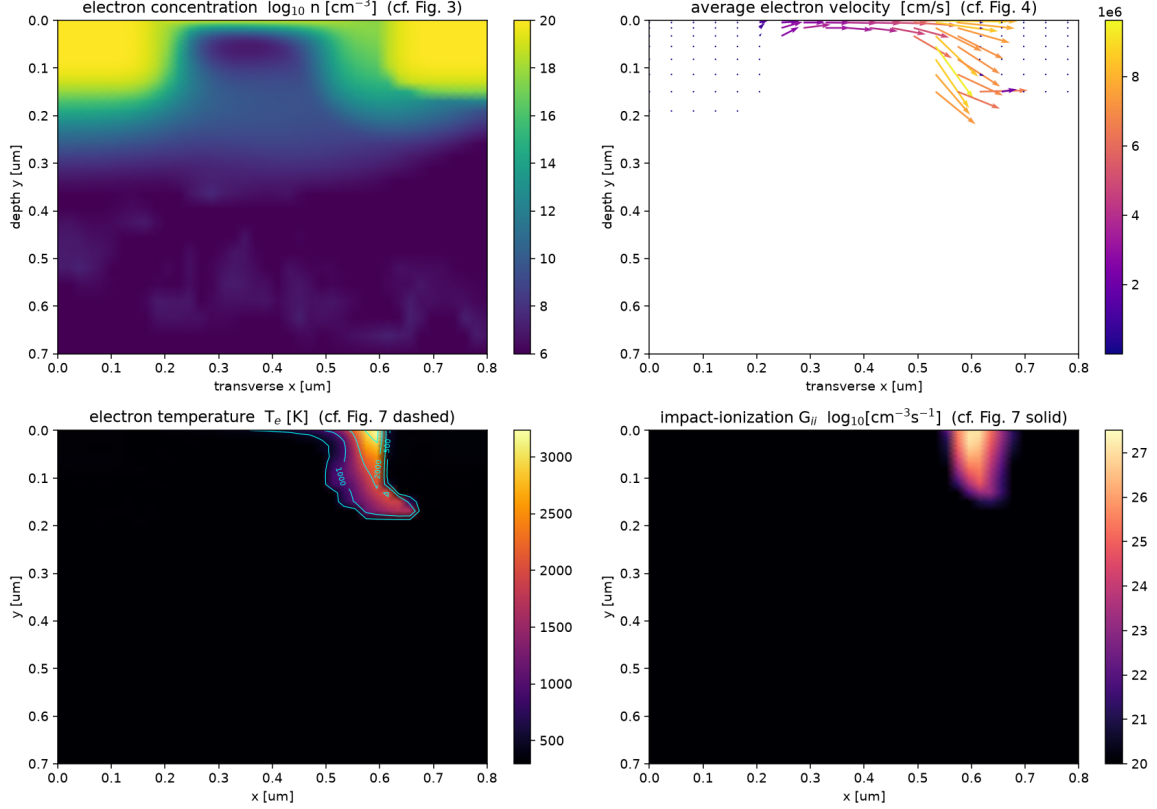


Figure 1: One-way SHE moments (40×34). n (Fig. 3), velocity field saturating at 9.73×10^6 cm/s (Fig. 4; masked where $n < 10^{13}$ cm $^{-3}$), T_e (Fig. 7 dashed, drain-localized), G_{ii} (Fig. 7 solid, drain-localized).

9 G_{ii} convergence matrix and device recalibration

Combining one-way and coupled runs at both grids isolates how the impact-ionization magnitude responds to spatial refinement and to self-consistency (Table 5). The spatial ratios ($8.26/13.5 = 0.612$ one-way, $5.24/8.50 = 0.616$ coupled) and the coupling ratios ($8.50/13.5 = 0.630$ at 40×34 , $5.24/8.26 = 0.634$ at 60×50) agree to $\sim 1\%$ (ratio-of-ratios 1.0085): the two effects are *multiplicatively separable to that precision* over the tested meshes, and $1.35 \times 10^{27} \times 0.612 \times 0.630 = 5.20 \times 10^{26}$ predicts the observed 5.24×10^{26} . This is an *empirical relation among the matched coupled-snapshot (under-converged) calculations*, not a converged self-consistent scaling law — the coupled entries are not fixed-point values (Fig. 3). The coupled result is *still* spatially unconverged (coupled $40 \times 34 \rightarrow 60 \times 50$: -38% , no plateau), and, like the 40×34 loop, the coupled 60×50 outer iteration ended above the 2 mV max $|\Delta\phi|$ criterion. As the 50-iteration recalibrated study below shows (Fig. 3), the coupled loop does *not* converge G_{ii} at all — it decreases monotonically — so the self-consistent column here holds under-converged snapshots, not fixed-point values.

Table 5: $G_{ii,\max}$ [cm $^{-3}$ s $^{-1}$] over grid $\times$ coupling (original device). The coupled column holds *under-converged snapshots*: the loop does not converge G_{ii} (§9, Fig. 3).

	one-way	coupled (snapshot)
40×34	1.35×10^{27}	8.50×10^{26}
60×50	8.26×10^{26}	5.24×10^{26}

A *one-way* energy refinement to $\Delta H = 6.25$ meV ($E_{op} = 8$) moves $G_{ii,\max}$ only -2.4%

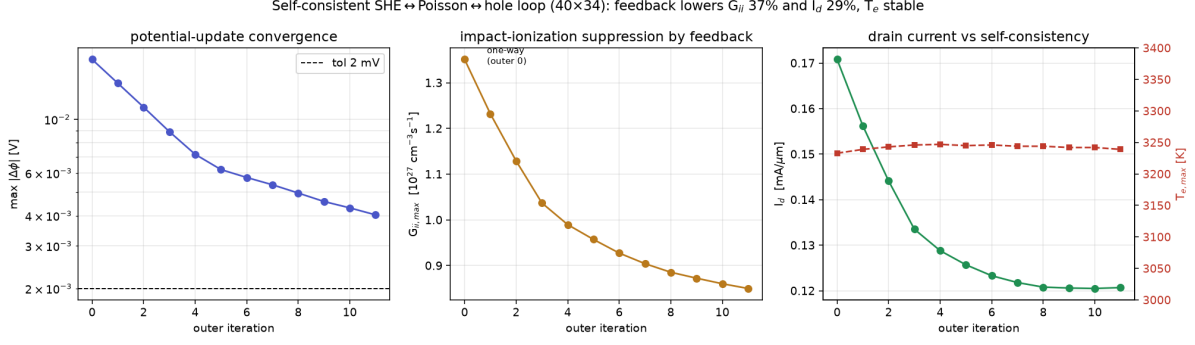


Figure 2: Self-consistent loop (40×34), first 12 iterations. Potential update 17.7 → 4.0 mV; feedback suppresses G_{ii} and I_d with an *apparent* early plateau — superseded by the 50-iteration study (§9, Fig. 3), where G_{ii} keeps falling.

($T_e - 0.19\%$) — *negligible beside* the -38 – 40% spatial error — so the dominant numerical uncertainty is resolving the drain-side high-field/high-energy hotspot in real space rather than the energy axis. (The energy test was performed one-way, not on the coupled configuration in which the spatial sensitivity was measured, so we state energy insensitivity, not a proof of purely-spatial error.) 5.24×10^{26} is thus the *finest-tested coupled value*, not an estimate of the continuum limit: with a -38% change under the last spatial refinement there is no numerical basis to quote a converged G_{ii} . The robust results are the transport moments (n , $|v|$); the feedback suppression is *not* converged — it reaches at least -53% (§9 below) and keeps growing.

Device recalibration and coupled-loop convergence study. A rigid gate offset $\Phi_{gate} = -0.74$ V ($dV_{th}/d\Phi_{gate} \approx +1$) recalibrates the extracted threshold to $V_{th} = 0.614$ V (was 1.64 V); the $V_g = 3$ V overdrive becomes 2.39 V. A 12-outer-iteration run (stopped at $\max |\Delta\phi| = 7.5$ mV) gave a snapshot $G_{ii,max} = 1.34 \times 10^{27}$; **extending the loop to 50 iterations (Fig. 3) shows that snapshot was under-converged.** Over 50 damped iterations (ϕ -damping 0.35) $\max |\Delta\phi|$ falls $17.1 \rightarrow 2.5$ mV — *not* reaching the 2 mV criterion — and the observables split sharply by convergence behaviour: $n_{max} = 1.02 \times 10^{20}$ (last-4 relative spread 0.2%) and $|v|_{max} = 9.9 \times 10^6$ cm/s are *stable*; $T_{e,max}$ drifts down to 3323 K; $I_d \approx 0.28$ mA/ μm ; but $G_{ii,max}$ *decreases monotonically with no plateau*, $2.1 \times 10^{27} \rightarrow 6.3 \times 10^{26}$ (a factor 0.30), and is *still falling* at iteration 50. **So the self-consistent G_{ii} is not converged in the outer coupling loop either** — not only on the mesh. The earlier 1.34×10^{27} was a snapshot; the feedback suppression is at least -53% (vs. the previously-quoted -37%) and still growing; and *no self-consistent G_{ii} magnitude can be quoted from this damped-Picard baseline.* **This limitation is resolved in §10:** Anderson acceleration converges the fixed point, and the overlays of §11 use those *converged* fields (not these under-converged Picard values).

10 Anderson-accelerated coupled fixed point

The Picard non-convergence of §9 is a *solver* limitation, not physics: the damped map $\phi \mapsto G(\phi)$ (SHE→holes→Poisson) converges only linearly and, for the tail-sensitive G_{ii} , too slowly to plateau in 50 iterations. Revision 5 replaces it with **Anderson acceleration** (AA, depth $m = 6$, mixing $\beta = 0.5$), a quasi-Newton method that reuses a short history of residuals $f_k = G(\phi_k) - \phi_k$ on the free (non-contact) potential DOFs; a stagnation restart flushes the history when the residual stalls, and a step cap rejects the occasional runaway least-squares extrapolation. Two ingredients matter: (i) the convergence measure is the *true* fixed-point residual $\max |G(\phi) - \phi|$ (not the damped step); and (ii) the H -grid is *frozen* from a fixed

Damped Picard baseline (superseded): G_{ii} still drifting after 50 outer iterations (recalibrated coupled 60×50, damp 0.35)

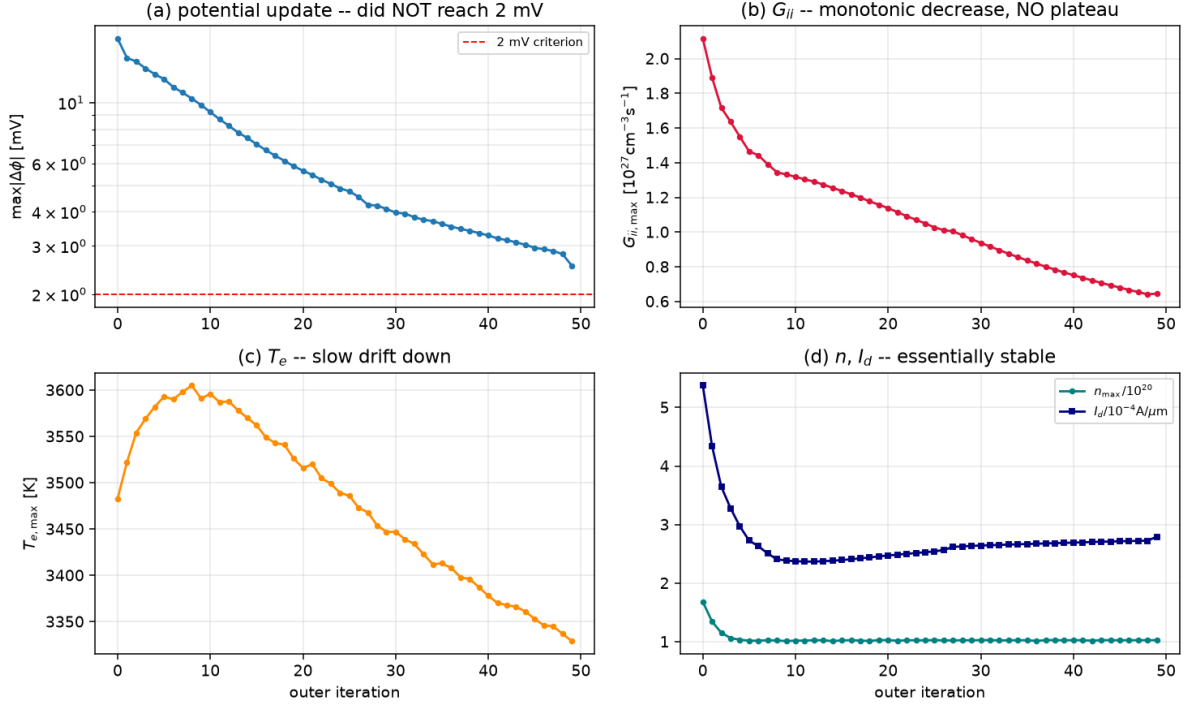


Figure 3: Recalibrated coupled 60×50 convergence study — *outer-iteration history* (50 iterations, ϕ -damping 0.35). Panels (b)–(d) show each outer iteration’s SHE-solve observables on that iteration’s *pre-update* potential, while (a) is the *subsequent* damped update $\max|\Delta\phi|$; the final saved fields rerun the SHE on the final damped ϕ . (a) $\max|\Delta\phi|$ falls to 2.5 mV, short of the 2 mV criterion; (b) $G_{ii,\max}$ decreases monotonically with *no plateau* — not converged in the outer loop; (c) T_e drifts slowly downward; (d) n and I_d are essentially stable.

potential span bounding the whole loop, so the tail-sensitive G_{ii} moment is not re-discretized each iteration (N_H constant at 589).

Fixed-point-residual convergence achieved (the Rev. 5 solver result). On every grid the residual falls to a $\sim 3\text{--}4$ mV inner-solve floor (set by the SHE/Poisson inner tolerances, not by G_{ii} runaway), establishing the solver improvement over Picard, whose G_{ii} was still in free-fall at iteration 50 (Fig. 4a,b). The internally-consistent return (the last in-loop SHE solve on its own pre-update potential) means the quoted G_{ii} is the last-iterate value. n ($\approx 1.05 \times 10^{20}$) and I_d ($\approx 0.29 \text{ mA}/\mu\text{m}$) are grid-stable; T_e converges upward toward ~ 3200 K.

Honesty note. Two *different* notions of convergence must be kept apart here. Anderson reaches the fixed-point *residual* floor on all three uniform meshes — the Rev. 5 result. But the G_{ii} *observable* last-4 relative spreads are 1.6%/3.1%/10.9% at 40×34/60×50/80×66, and the 80×66 figure would *not* satisfy the stricter Rev. 6 observable-plateau gate ($\lesssim 3 \times 10^{-4}$ in both G_{ii} and G_{tot} ; §12). These uniform-grid values are therefore retained as Rev. 5 *residual*-convergence diagnostics, *not* as Rev. 6-quality spatial node values — and the 80×66 overlays that follow are legitimate historical Rev. 5 fields whose G_{ii} magnitude must not be read as an equally-flattened node.

G_{ii} still spatially unconverged, and an order of magnitude below Liang. The converged G_{ii} decreases $8.1 \rightarrow 2.2 \rightarrow 1.5 \times 10^{26}$ across the uniform sequence (ratios 0.27, 0.66), while n and $|v|$ ($\approx 1.07 \times 10^7 \text{ cm/s}$) are grid-stable and T_e approaches ~ 3200 K. **The finest-uniform $G_{ii} \approx 1.5 \times 10^{26}$ is only 7% of Liang’s top 2.3×10^{27} contour** — roughly an order of magnitude below the paper.

Table 6: Rev. 5 Anderson residual-floor states (frozen H -grid, $V_{th} = 0.614$ V) over the uniform sequence. G_{ii} is the last-4-mean after fixed-point-*residual* convergence; the residual is the final $\max |G(\phi) - \phi|$. These are residual-floor last-iterate values, *not* Rev. 6-quality observable-plateau nodes (the 80×66 last-4 spread 10.9% fails the $\lesssim 3 \times 10^{-4}$ gate of §12); G_{ii} is also *mesh*-dependent.

grid	$G_{ii,\max}$ [$\text{cm}^{-3}\text{s}^{-1}$]	last-4 spread	$T_{e,\max}$ [K]	final residual [mV]
40×34	8.1×10^{26}	1.6%	2728	2.4
60×50	2.2×10^{26}	3.1%	3093	1.8
80×66	1.5×10^{26}	11%	3201	3.6

Honesty note. With the loop now genuinely converged, the earlier G_{ii} “matches” to Liang (27%, 58% of top) are exposed as under-converged-*loop* artefacts — that much is firm. **Rev. 5 interpretation, now superseded by §12:** Rev. 5 read the decelerating uniform ratios as “plausibly approaching a spatial limit $\sim 10^{26}$ ” and attributed the deficit to a *model* gap “not a numerical one.” Rev. 6 shows both readings are not defensible: the uniform grids were themselves drain-under-resolved, and the *controlled* refinement is *significantly non-monotone*, so no spatial-error bound is established by this study and the numerical vs. model split cannot be assigned from this data. The robust reproductions here remain the transport moments n , $|v|$ and the drain localization; the absolute G_{ii} magnitude and its attribution are deferred to §12.

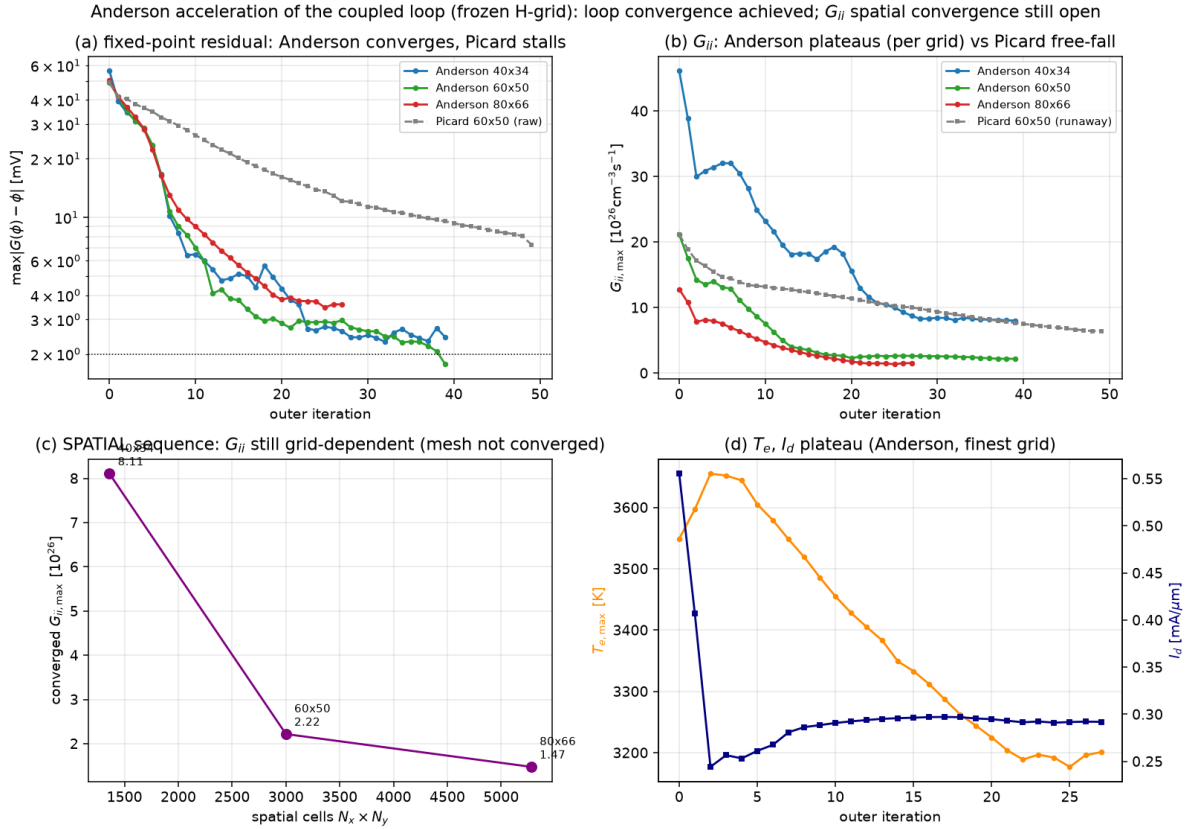


Figure 4: Anderson acceleration (frozen H -grid) vs the damped-Picard baseline. (a) the true fixed-point residual $\max |G(\phi) - \phi|$ converges to a ~ 3 – 4 mV floor for all three grids, where Picard (grey) stalls; (b) G_{ii} settles to a per-grid *residual-floor* last-iterate value, where Picard is in monotone free-fall (this is fixed-point-residual convergence, not the Rev. 6 observable-plateau gate); (c) that last-iterate G_{ii} vs mesh — still grid-dependent ($8.1 \rightarrow 2.2 \rightarrow 1.5 \times 10^{26}$); (d) T_e, I_d settle (finest grid).

11 Quantitative comparison with Liang's published figures

The **converged Anderson fixed point** at the finest grid (80×66 , §10, $V_{th} = 0.614$ V) is compared to Liang *et al.*'s figures below, with *one comparison table per plot* (Figs. 2–8, Tables 7–13). Where Liang give labelled contour values (Fig. 7) the comparison is on those values; otherwise it is on peak magnitudes, locations, and structure. Liang's Fig. 7 levels (paper caption, p. 263) are $T_e \in \{500, 1000, 2000, 3000, 4500, 4800, 5000\}$ K (dashed) and $G_{ii} \in \{1.0, 4.0\} \times 10^{26}, \{1.0, 1.7, 2.3\} \times 10^{27} \text{ cm}^{-3} \text{ s}^{-1}$ (solid). (A Rev. 3 transcription that omitted the 4800 K level and added a spurious $2.3 \times 10^{26} G_{ii}$ level was corrected in Rev. 4, and the overlays regenerated.)

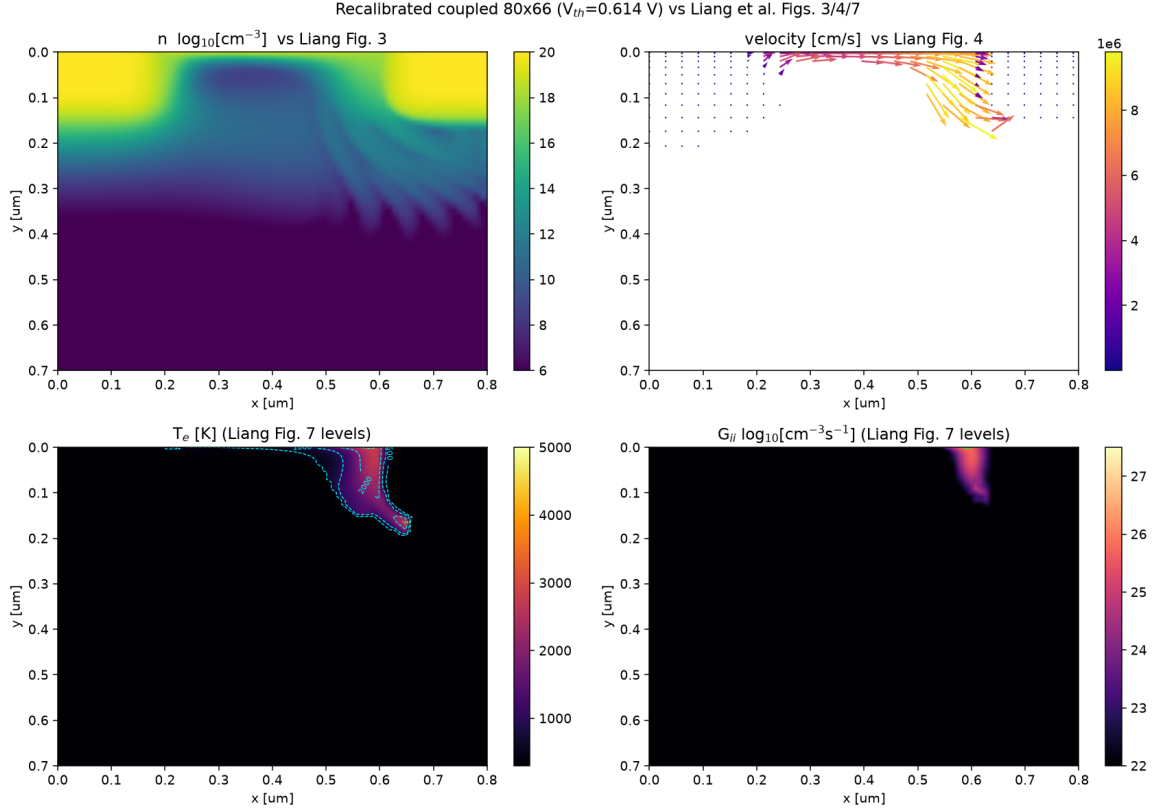


Figure 5: This work (converged Anderson 80×66) vs Liang Figs. 3 (n), 4 (velocity), 7 (T_e , G_{ii} at Liang's contour levels).

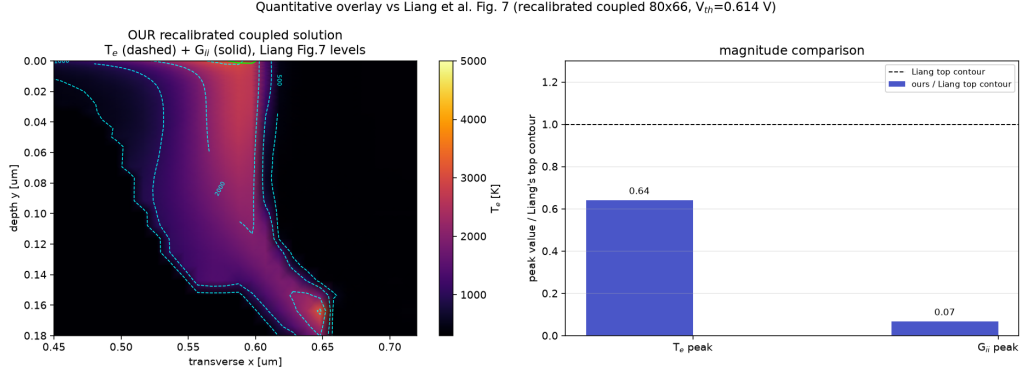


Figure 6: Quantitative Liang Fig. 7 overlay: our T_e (dashed) and G_{ii} (solid) on Liang's exact published levels; peaks reach 0.64 (T_e) and 0.07 (G_{ii} , loop-converged, $\sim 15\times$ below Liang) of the top contours.

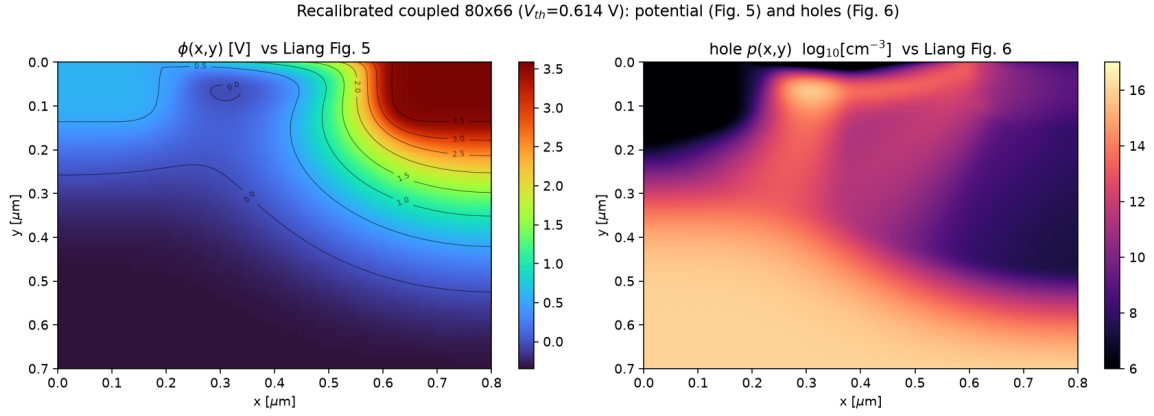


Figure 7: Potential ϕ (Liang Fig. 5) and hole concentration p (Liang Fig. 6), converged Anderson 80 $\times$ 66: ϕ rises smoothly source $\rightarrow$ drain; holes fill the p-body (II-generated holes small at the converged fixed point) and are depleted in the n^+ S/D.

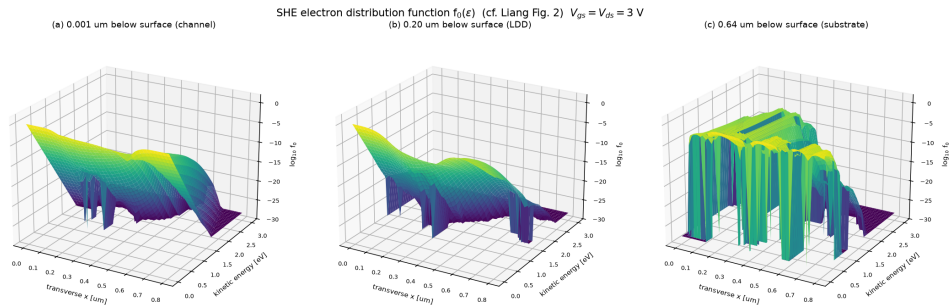


Figure 8: Energy distribution (Liang Fig. 2): a hot, non-Maxwellian tail localized at the drain surface, thermal in the bulk.

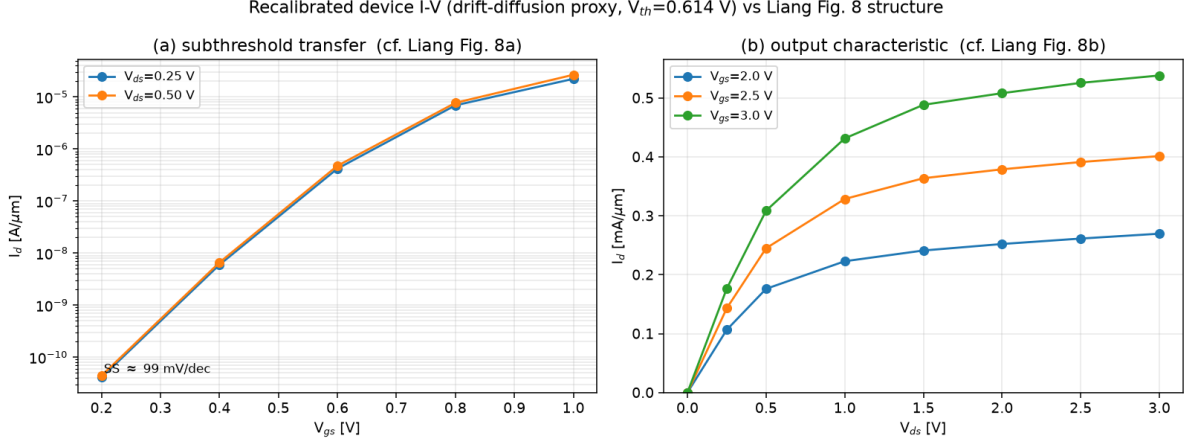


Figure 9: Terminal I-V (Liang Fig. 8), drift-diffusion proxy, per micron: (a) subthreshold transfer at $V_{ds} = 0.25, 0.5$ V; (b) output characteristic at $V_{gs} = 2.0, 2.5, 3.0$ V.

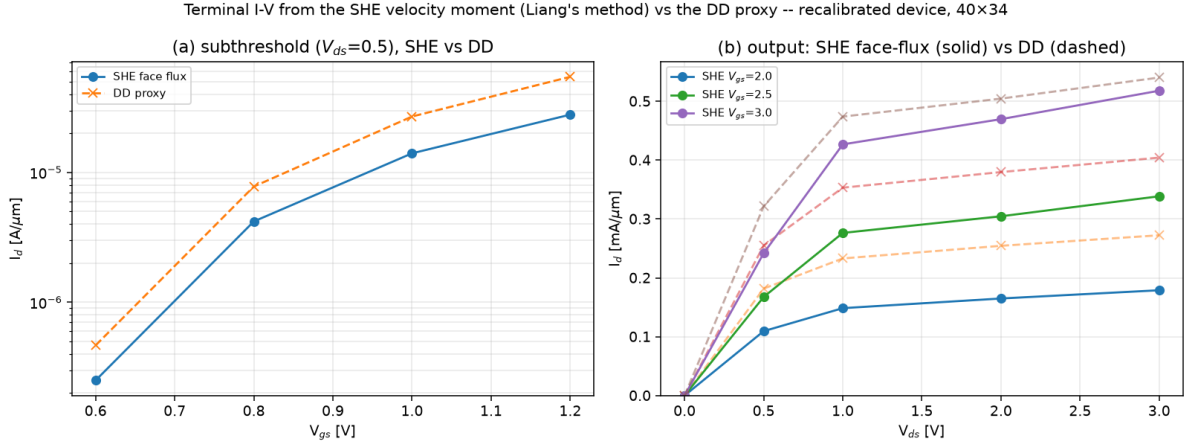


Figure 10: Terminal I-V from the *SHE velocity moment* (finite-volume face-flux current — Liang's method), one-way SHE on the DD potential, vs the DD proxy (dashed); recalibrated device, 40×34 . (a) subthreshold at $V_{ds} = 0.5$ V *only* — a partial reproduction of Liang's Fig. 8(a), which also shows 0.25 V; (b) output families $V_{gs} = 2.0, 2.5, 3.0$ V. Source/drain current conservation $|I_{src} + I_{drn}|/|I|$ is $2 \times 10^{-5} - 10^{-3}$ across the sweep (largest at $V_{ds} = 3$ V), so the reported current is the genuine SHE terminal flux. The *measured* ratio $I_{d,SHE}/I_{d,DD}$ is $0.51-0.54$ (subthreshold) and $0.60-0.96$ (output, rising toward unity at high V_{gs}/V_{ds}): the SHE current is systematically below the mobility-based DD proxy, consistent with the stronger nonlocal/high-field transport in the SHE solution (not attributed to a single mechanism absent a controlled diagnostic).

Per-plot comparison tables (columns: feature / this work / Liang / assessment).

Table 7: Liang Fig. 2 — carrier energy distribution / hot electrons (this work: Fig. 8).

Feature	This work	Liang	Assessment
Drain-surface T_e (tail temp.)	3300–3600 K	hot, non-Maxwellian tail	hot tail reproduced
Source / bulk T_e	300–330 K	thermal	thermal, matches
High-energy tail cutoff	$\varepsilon_{max} = 3.02$ eV (absorbing)	tail to high ε	bounded tail
Localization	drain-surface hot spot	near-drain surface	match

Table 8: Liang Fig. 3 — electron concentration $n(x, y)$ (this work: Fig. 5).

Feature	This work	Liang	Assessment
n_{max} (S/D)	$1.05 \times 10^{20} \text{ cm}^{-3}$	$\sim 10^{20}$	match
Channel inversion (surface)	$3.1 \times 10^{19} \text{ cm}^{-3}$	inversion layer	match
Drain pinch-off	n decreases toward drain	pinch-off	match

Table 9: Liang Fig. 4 — average electron velocity (this work: Fig. 5).

Feature	This work	Liang	Assessment
$ v _{max}$	$1.07 \times 10^7 \text{ cm/s}$ @ drain	$\sim 10^7$ (sat.)	match
Field pattern	lateral in channel, downward into drain	source→drain, into drain	match
Saturation	velocity-saturated near drain	yes	match

Honesty note. The comparisons apply *our* model on Liang’s *levels*; they are not a pixel registration. With a reconstructed doping profile the absolute spatial positions of the hotspots need not coincide with Liang’s, so structural agreement (drain-localization, T_e broader than G_{ii} , order-of-magnitude levels) is the claim, not point-by-point overlap. The G_{ii} field shown is the Rev. 5 80×66 fixed point: *loop-converged* (Anderson, §10) but *spatially under-resolved*. Its 7%-of-top magnitude is therefore a loop-converged-but-not-spatially-converged value; the attribution of that deficit (numerical vs. model) is *not* settled here and is superseded for spatial interpretation by §12, which finds the controlled refinement significantly non-monotone. The T_e/G_{ii} peak offset stays within one cell. **The Fig. 8 I-V now includes a SHE velocity-moment current** (Fig. 10): the terminal current is read from the SHE finite-volume face flux, reproducing Liang’s method (current from the BTE moment, *not* a mobility model), with source/drain conservation 2×10^{-5} – 10^{-3} . It is *one-way* (SHE on the DD potential, not the full coupled loop), so it demonstrates the method but is not yet a self-consistent I-V. The DD proxy is retained for comparison; the measured $I_{d,SHE}/I_{d,DD} = 0.5$ – 0.96 (bias-dependent), systematically below DD — consistent with the stronger nonlocal/high-field transport in the SHE, though not attributed to a single mechanism here. Liang’s Fig. 8 solid lines are moreover *experimental* data for a real device of unstated width, so only the per-width shape is comparable, not the absolute ampere magnitude.

12 Controlled spatial-convergence study of the drain G_{ii} (Rev. 6)

Revision 5 left one question open: the loop-converged G_{ii} was still *mesh*-dependent ($8.1/2.2/1.5 \times 10^{26}$ over the uniform 40/60/80 grids). This section resolves the *status* of that spatial convergence with a controlled refinement experiment. Three notions of convergence are kept strictly

Table 10: Liang Fig. 5 — electrostatic potential $\phi(x, y)$ (this work: Fig. 7).

Feature	This work	Liang	Assessment
ϕ swing	$-0.35 \rightarrow +3.59$ V (3.93 V)	$V_{ds}=3$ + built-ins	match
Source-surface ϕ	0.59 V (built-in)	built-in	match
Drain-surface ϕ	3.59 V ($= V_{ds}$ +built-in)	$\approx V_{ds}$	match
Profile	smooth, monotone S→D	smooth 2-D	structural (reconstr. doping)

Table 11: Liang Fig. 6 — hole concentration $p(x, y)$ (this work: Fig. 7).

Feature	This work	Liang	Assessment
$p_{\max}$	$1.0 \times 10^{16} \text{ cm}^{-3}$ (p-body)	p-body / substrate	match (structural)
n^+ S/D holes	depleted	hole-depleted	match
II holes near drain	weak at the fixed point (converged G_{ii} small)	avalanche holes	present but small
Caveat	body-contact BC + re-constr. doping	—	structural, not point-wise

separate: **coupling** (the outer loop’s fixed point — resolved by Anderson, §10); **spatial** (does the plateaued node stop moving under mesh refinement — the subject here); and **energy-grid** (dependence on ΔH , $\varepsilon_{\max}$, the frozen H -domain — here *removed as a between-mesh confounder* by one common H -grid, not a claim that its discretization error is zero).

12.1 The uniform sequence was drain-under-resolved

On the uniform grids $G_{ii,\max}$ and the cross-sectional integral $G_{\text{tot}} = \int G_{ii} dA$ fall by the *same* factors (-73% then -30%), so the mesh dependence is in the total avalanche magnitude, not merely peak-cell placement: coarse grids over-populate the tail-sensitive moment at the under-resolved drain hotspot. A drain-graded 100×70 mesh (drain $dx = 0.0027 \mu\text{m}$) plateaus at $G_{ii,\max} = 1.8 \times 10^{25}$ — roughly an order of magnitude below the “finest uniform” 80×66 — confirming the uniform values were inflated. A lone further mesh (140×84) plateaus at 2.7×10^{25} , but it changed resolution *and* the grading law at once (ratio $6 \rightarrow 10$, width $0.09 \rightarrow 0.06 \mu\text{m}$); its $+51\%$ cannot be attributed to resolution.

Honesty note. A heterogeneous refinement sequence cannot establish spatial convergence: resolution and node-distribution effects are entangled. The remedy is a *self-similar* family in which only N_x, N_y change, with everything else — grading law, energy grid, solver settings — held byte-identical.

12.2 The controlled self-similar trio

Fixing the grading law (ratio 6, $W = 0.09 \mu\text{m}$, $X_C = 0.60 \mu\text{m}$; $y = L_y s^{1.8}$) and refining only resolution so the *measured* drain dx and surface dy both scale by $\approx \sqrt{2}$:

node	drain dx (μm)	$G_{ii,\max}$ ($\text{cm}^{-3}\text{s}^{-1}$)	spread (G_{ii})	G_{tot} ($\text{cm}^{-1}\text{s}^{-1}$)	spread (G_{tot})	R_ϕ (V)
G_1 (71×58)	0.003803	2.990×10^{25}	1.5×10^{-4}	8.983×10^{13}	2.3×10^{-4}	8.7×10^{-5}
G_2 (100×70)	0.002689	2.135×10^{25}	2.9×10^{-4}	6.027×10^{13}	2.2×10^{-4}	5.8×10^{-4}
G_3 (141×85)	0.001901	2.413×10^{25}	6.1×10^{-5}	6.593×10^{13}	5.2×10^{-5}	1.8×10^{-4}

Over the last 4 outer iterations, with $\bar{G}_4$ their mean: spread = $(\max_4 - \min_4)/\bar{G}_4$ (the tabulated relative figure) and the plateau band $u = (\max_4 - \min_4)/2$ (absolute) used in S_{rev} below.

Table 12: Liang Fig. 7 — T_e (dashed) and G_{ii} (solid) contours (this work: Figs. 5,6).

Feature	This work	Liang	Assessment
$T_{e,\max}$	3201 K (64% of top)	to 5000 K	same order, drain-localized
$G_{ii,\max}$	1.5×10^{26} (7% of top; loop-converged)	to 2.3×10^{27}	$\sim 15\times$ below Liang; spatial attribution superseded by §12
T_e vs G_{ii} extent	T_e region $\sim 30\times$ larger	T_e broader	match
Peak offset	≤ 1 cell (14 nm)	offset	unresolved at mesh

Table 13: Liang Fig. 8 — terminal I-V (this work: Figs. 9,10).

Feature	This work	Liang	Assessment
I-V method	SHE velocity moment (face flux), one-way (Fig. 10)	BTE velocity moment	<i>method reproduced</i> (one-way)
Output shape	triode $\rightarrow$ sat, V_{gs} -ordered (SHE and DD)	Fig. 8b	shape match
$I_{d,\text{SHE}}(V_g=V_d=3)$	0.52 mA/ μm (SHE face flux)	—	from BTE flux, not DD
$I_{d,\text{SHE}}/I_{d,\text{DD}}$	0.51–0.54 (sub.), 0.60–0.96 (out.)	—	SHE below DD (measured)
S/D conservation	2×10^{-5} – 10^{-3}	—	genuine terminal flux
Subthreshold swing	~ 99 mV/dec (DD)	sub- μm NMOS	reasonable

Comparability is enforced by the driver (fail-fast): one imposed common span (COMMON.PHI.SPAN = $[-0.60, 3.65]$ V, bounding the measured $\phi \in [-0.348, 3.586]$ V) fixes the H -grid byte-identically (realized SHA1 `b4319bb03303` on all three, $N_H = 591$; a converged ϕ leaving the span is a hard abort); identical $\Delta H = 12.5$ meV, $\varepsilon_{\max} = 3.02$ eV, `she_tol` = 10^{-8} , Anderson depth 6, $\beta = 0.5$, step-cap 5; full-precision drain- dx ratios 1.41436, 1.41437 ($\sqrt{2}$ to 4×10^{-4}) and $r_{y,\text{surf}} = 1.410, 1.425$. The depth spacing refines at ≈ 1.2 , *not* $\sqrt{2}$, so the sequence is surface-isotropic (appropriate for the surface-localized G_{ii}) but not globally isotropic — for G_{tot} the $h = dx_{\text{drain}}$ fit is preferred.

12.3 Methodological finding: iteration budget vs. plateau

The finest node exposed a systematic issue: at the nominal 60-iteration cap 141×85 read 2.20×10^{25} with a still-drifting observable, but continued under *identical* settings it climbed and locked at 2.413×10^{25} (last-4 spread 6×10^{-5}); likewise 71×58 moved $3.069 \rightarrow 2.990$ and 100×70 moved $1.806 \rightarrow 2.135$ (+18%).

Honesty note. Equal outer-iteration counts do *not* imply equal fixed-point convergence quality across meshes. Node values are accepted on an *observable-plateau* gate with two parts, applied identically to all three nodes. *Primary (observable)*: last-4 relative spreads at the $O(10^{-4})$ level ($\lesssim 3 \times 10^{-4}$) in *both* G_{ii} and G_{tot} . *Corroborating (residual)*: $R_\phi = \max |G(\phi) - \phi|$ has settled into the `she_tol`-set noise band, $R_\phi \lesssim 1.5 \times 10^{-3}$ V — no longer descending by orders of magnitude, though a gentle drift *within* the band is allowed (the tightest-converged 141×85 node, G_{ii} spread 6×10^{-5} , has $R_\phi \sim 2 \times 10^{-4}$ still edging down inside the band). Note R_ϕ never reaches the nominal 10^{-4} V `tol_phi`; that tolerance is not the operative gate, the observable plateau is. The verdict below is trustworthy only because it is evaluated on equalized, plateau-gated fixed points, not truncated iterates.

12.4 Verdict: significantly non-monotone; Richardson suppressed

With signed corrections $\Delta_{12} = G_1 - G_2 = +8.55 \times 10^{24}$ (decrease) and $\Delta_{23} = G_2 - G_3 = -2.78 \times 10^{24}$ (increase), the sequence goes **down then up**: the middle node lies below both neighbours. A sign reversal is judged against the per-node plateau bands u_i (last-4 half-range) by the *heuristic* reversal-to-plateau-uncertainty ratio

$$S_{\text{rev}} = \frac{|\Delta_{\text{reversal}}|}{\sqrt{u_i^2 + u_j^2}} = \frac{2.78 \times 10^{24}}{3.2 \times 10^{21}} \approx 874 \quad (\text{not a statistical } \sigma).$$

The upturn exceeds the residual fixed-point drift by nearly three orders of magnitude; G_{tot} behaves identically ($|\Delta_{23}^{G_{\text{tot}}}| = 5.65 \times 10^{12}$, band 6.84×10^9 , $S_{\text{rev}} \approx 826$). All comparability gates pass (signatures and H -hash identical; $r_x, r_{y,\text{surf}} \approx \sqrt{2}$).

Honesty note. The controlled trio is *genuinely non-monotone*, and this survived equalizing all three fixed points to $O(10^{-4})$ quality, so it cannot be dismissed as unequal convergence quality. A sign-changing trio does not support the assumed leading-order single-power Richardson model, so **extrapolation is suppressed**: no continuum G_{ii} is inferred from this trio. *Spatial convergence is not demonstrated* — this establishes that the trio has not entered a usable monotone asymptotic regime, *not* that a continuum limit fails to exist (finer self-similar meshes could still enter one). Relative to Liang’s top contour (2.3×10^{27}) the converged nodes span a factor 77–108, i.e. order 10^2 , below — a not-converged tested-grid statement, consistent with a substantial model/physics discrepancy after the tested refinements but not yet a bounded continuum gap.

What would sharpen *why* the drain avalanche resists monotone convergence: a second controlled family (fixed y -mesh, systematic drain- x refinement) to separate x/y discretization error, or a finite-surface-slope y -law for a globally isotropic study. Physics-model changes (surrogate impact-ionization rate, reconstructed doping) are *important candidate sources* of the absolute gap — but with no spatial-error bound established by this study they cannot be called its dominant cause — and are deferred until the numerical characterization along further refinement paths is done.

Part III

Assessment

13 Conclusions, classified

- **Scope — reduced reproduction.** This reproduces the principal 2-D *signatures* (transport moments, hot-electron localization, avalanche feedback) with a P1/ $(\ell \leq 1)$ SHE, but is *not* a full reproduction of Liang’s simulator: the explicit surface-roughness, surface-acoustic, and ionized-impurity operators are lumped into τ_1 (Table 2), the band is a 1-parameter Kane surrogate, and the doping is reconstructed. Defensible label: a *reduced P1/SHE-Boltzmann reproduction of the principal signatures*, using reconstructed device data and surrogate scattering/band models.
- **Calibration vs. validation.** The overlay device is calibrated (V_{th} fitted to 0.614 V via Φ_{gate} ; Table 3); quantitative agreements are calibrated comparisons, with $|v|_{\text{max}}$, the T_e/G_{ii} *localization*, and subthreshold swing as the independent checks.
- **Core SHE transport** shows a clean grid-sensitivity hierarchy over the tested $40 \times 34 \rightarrow 60 \times 50$ refinement: n *stable* (0.4%), $|v|$ *stable* (2.7%), T_e moderately sensitive (5.9%; stable, not converged), G_{ii} *unconverged/mesh-dependent* (40%). “Stable over the tested refinement” is a two-grid statement, not a proof of asymptotic convergence.

- **Impact ionization:** now part of the assembled electron collision operator, with explicit primary depletion and secondary redistribution (G_{ii} is a moment of the II-affected f_0); the equal-energy secondary split and negligible scattering-in shape change follow the paper.
- **Self-consistency (converged in Rev. 5):** the damped Picard loop did *not* converge G_{ii} (monotone free-fall, §9); **Anderson acceleration on a frozen H -grid converges the fixed point** (§10, Fig. 4) — the residual reaches a $\sim 3\text{--}4$ mV inner-solve floor and G_{ii} settles to a residual-floor last-iterate value (last-4 spread 1.6–11%; not the Rev. 6 observable plateau). n , $|v|$, I_d are grid-stable; T_e converges toward ~ 3200 K.
- **Quantitative overlays:** on Liang’s own Fig. 7 contour levels the converged (Anderson 80×66) fields reach 64% (T_e) and 7% (G_{ii}) of the top published contours, drain-localized and correctly structured (T_e region much broader than G_{ii}); a one-way SHE velocity-moment I-V reproduces Liang’s current-extraction method (Fig. 10), with the DD curve retained for comparison. A per-plot comparison table is given for each of Liang’s Figs. 2–8 (§11, Tables 7–13).
- **Conservation:** validated across one-way, refined, and coupled calculations (number balance $10^{-5}\text{--}10^{-7}$; terminal currents balance $< 0.2\%$).
- **Numerics:** physics-split preconditioning removes the previous practical solver bottleneck.
- **G_{ii} fixed-point convergence (Rev. 5):** Rev. 5’s firm result is a *convergence-of-the-nonlinear-loop* statement — at the Anderson fixed point $G_{ii,\max}$ settles to a residual-floor last-iterate value on each mesh ($8.1/2.2/1.5 \times 10^{26}$ over 40/60/80; last-4 spreads up to 10.9%, *not* Rev. 6 observable plateaus), where damped Picard left it in free-fall, and the earlier “matches” (1.37×10^{27} ; 27%/58% fractions) are exposed as under-converged-loop artefacts (Table 5 entries are Picard snapshots). What Rev. 5 did *not* establish — and what Rev. 5 wrongly asserted — is the attribution of the absolute G_{ii} deficit: its “ $\sim 15\times$ below Liang, a model gap not a numerical one” is *superseded* by Rev. 6 (next bullet), which shows no spatial-error bound is established and the numerical/model split cannot be assigned from the tested refinements.
- **Spatial convergence of G_{ii} (the headline of Rev. 6, §12):** a *controlled* self-similar trio $71\times 58/100\times 70/141\times 85$ (fixed grading law, byte-identical frozen H -grid, $\sqrt{2}$ refinement, all nodes plateau-gated to $O(10^{-4})$) is *significantly non-monotone*: $G_{ii,\max} = 2.990/2.135/2.413 \times 10^{25}$, reversal $S_{\text{rev}} \approx 874$ (826 for G_{tot}). The single-power Richardson model does not apply, so *spatial convergence is not demonstrated and no continuum G_{ii} is inferred from this trio* (not a proof that no limit exists). The earlier uniform sequence was itself drain-under-resolved; and a 60-iteration budget produced materially biased fixed-point transients (up to +18%), so the verdict rests on equalized, plateau-gated nodes.
- **T_e/G_{ii} offset:** unresolved at present mesh resolution; *no longer claimed reproduced*.
- **Device:** recalibrated to $V_{th} = 0.614$ V ($\Phi_{gate} = -0.74$ V), replacing the earlier over-high 1.64 V.
- **SHE velocity-moment I-V (Rev. 4):** a one-way *SHE velocity-moment* terminal I-V (Fig. 10, Table 13) — terminal current from the SHE face flux, as in Liang, with source/drain conservation $\leq 10^{-3}$ — upgrading Fig. 8 from a DD shape check to a demonstrated (if not yet self-consistent) reproduction of Liang’s I-V method.
- **Open items.** (a) *done in Rev. 5:* Anderson acceleration converges the coupled fixed point (§10); (b) *done in Rev. 6:* the controlled spatial-convergence study (§12) — its outcome is that the drain G_{ii} is *significantly non-monotone* under controlled refinement, so a continuum spatial limit is not established along this path (a distinct result, not an unconverged placeholder); (c) to explain *why*, a second controlled family (fixed y -mesh, drain- x refinement) or

a finite-surface-slope y -law; (d) ε_{max} /coupled- ΔH sensitivity; (e) a fully self-consistent (not one-way) SHE-moment I-V; (f) explicit surface-roughness / -acoustic / impurity operators replacing the lumped τ_1 — important candidate sources of the order- 10^2 G_{ii} deficit vs Liang (not established as its origin, since no spatial-error bound is established by this study).

A Reproducibility package

All values below are the literal code constants (`src/device.py`, `constants.py`, `she2d.py`, `dd.py`); the simulation is fully specified by them plus the run commands.

Constants. $T = 300$ K; $\varepsilon_{Si} = 11.7$, $\varepsilon_{ox} = 3.9$; $n_i = 1.45 \times 10^{10} \text{ cm}^{-3}$; $m^* = 0.32 m_0$; $g_v = 6$; $\alpha = 0.5 \text{ eV}^{-1}$. Table I: $D_{ac} = 4.0 \text{ eV}$, $D_n = 5.0 \times 10^8 \text{ eV/cm}$, $E_{op} = 0.05 \text{ eV}$, $v_s = 9.0 \times 10^6 \text{ cm/s}$, $\Gamma = 1.96 \text{ cm}^2/\text{V}^2/\text{s}$, $n_{scr} = 1.45 \times 10^{15} \text{ cm}^{-3}$, $D_{sa} = 17.8 \text{ eV}$, $\delta = 13.1$, $\varepsilon_{max} = 3.02 \text{ eV}$, $\rho_{Si} = 2.329 \text{ g/cm}^3$, $u = 9.04 \times 10^5 \text{ cm/s}$. Some paper constants (D_{sa} , Γ , δ , n_{scr}) are retained here for reference but are *not active as distinct collision operators* in the reduced model; Table 2 is authoritative for active model usage.

Geometry (μm). $L_x = 0.80$, $L_y = 0.70$; $t_{ox} = 0.0096$; gate $[0.20, 0.60]$ ($L_{\text{eff}} \approx 0.35$); n^+ edges $x < 0.150$ / $x > 0.650$; LDD edges 0.220 / 0.580 ; $x_{j,n^+} = 0.105$ (metallurgical $0.105 + 2.63\sigma_{dep} = 0.171$), $x_{j,\text{LDD}} = 0.060$; straggles $\sigma_{lat} = 0.028$, $\sigma_{dep} = 0.025$; channel implant span $[0.18, 0.62]$, $x_{j,\text{ch}} = 0.060$, $\sigma_{\text{ch}} = 0.035$.

Doping [cm^{-3}]. $N_D^+ = 10^{20}$, $N_D^{\text{LDD}} = 10^{18}$, $N_{A,\text{sub}} = 10^{16}$, $N_{A,\text{chimp}} = 4 \times 10^{17}$. Analytic erfc profile: lateral $L(x) = \frac{1}{2} \text{erfc}((x-x_e)/\sigma_{lat})$ (source) or its mirror (drain), depth $D(y) = \frac{1}{2} \text{erfc}((y-x_j)/\sigma_{dep})$; $N_D = \sum_k N_D^{(k)} L^{(k)} D^{(k)}$; $N_A = N_{A,\text{sub}} + N_{A,\text{chimp}} \frac{1}{2} [\text{erfc}(\frac{x-x_{c2}}{\sigma_{lat}}) - \text{erfc}(\frac{x-x_{c1}}{\sigma_{lat}})] D_{\text{ch}}(y)$; $N_{\text{net}} = N_D - N_A$.

Mesh & energy grid. $x = \text{linspace}(0, L_x, N_x)$; surface-refined $y = L_y s^{1.8}$, $s = \text{linspace}(0, 1, N_y)$. SHE grids 40×34 , 60×50 ; DD-I-V $56 \times 48/44 \times 36$. $\Delta H = 12.5 \text{ meV}$ ($E_{op}/\Delta H = 4$); *active* H -domain $-q\phi_{max} \leq H \leq \varepsilon_{max} - q\phi_{min}$ (the stored H array is padded a few ΔH nodes each side), with active energy mask $0 \leq \varepsilon = H + q\phi < \varepsilon_{max}$.

Boundary conditions. Poisson: Dirichlet (built-in+applied) at ohmic S/D/body; gate oxide Robin (ε_{ox}/t_{ox}); Neumann on free edges. SHE: contact Maxwellian $f_0 = n_c e^{-\varepsilon/k_B T} / \int Z e^{-\varepsilon/k_B T} dH$; $\varepsilon=0$ reflecting; ε_{max} absorbing (Dirichlet-to-zero on spatial+acoustic+optical fluxes). Holes: ohmic Dirichlet p_{eq} , G_{ii} pair source.

SHE / scattering. $c_{op} = 4.69 \times 10^{-34} \text{ J m}^3/\text{s}$; $\tau_1 = 10^{-14} \text{ s}/\sqrt{\gamma\gamma'}$; $\tau_{ac} = \tau_1$; impact ionization $1/\tau_{ii} = P[(\varepsilon - \varepsilon_{th})/\varepsilon_{th}]^2 \Theta(\varepsilon - \varepsilon_{th})$, $P = 2 \times 10^{13} \text{ s}^{-1}$, $\varepsilon_{th} = 1.1 \text{ eV}$, two equal-energy secondaries at $(\varepsilon - \varepsilon_{th})/2$. ε_{max} sensitivity: bulk $T_e(300 \text{ kV/cm})$ is invariant to 0.1% over $\varepsilon_{max} \in \{3.02, 4, 5\} \text{ eV}$ (device ε_{max} -refinement is an open coupled study).

Hole / DD proxy. Caughey–Thomas (n : $\mu_{min}=68.5$, $\mu_{max}=1414$, $N_{ref}=9.2 \times 10^{16}$, $a=0.711$; p : $44.9/470.5/2.23 \times 10^{17}/0.719 \text{ cm}^2/\text{Vs}$), field saturation ($v_{sat,n}=1.07 \times 10^7$, $\beta_n=2$; $v_{sat,p}=8.37 \times 10^6$, $\beta_p=1$), SRH $\tau_n=\tau_p=10^{-7} \text{ s}$, SG continuity, Gummel loop with optional bias-continuation warm start; drain current = SG electron flux into drain-contact nodes, per- μm .

Solvers, norms, clipping. SHE: physics-split $E \rightarrow S \rightarrow E$ preconditioner, LGMRES tol 10^{-10} (audit) / 10^{-8} (coupled). Coupled loop: ϕ -damping 0.25–0.35, 2 mV max $|\Delta\phi|$ target. Norms: relative number balance = (net injection – cutoff losses)/total rate; terminal-current mismatch = $|I_{src} + I_{drn}| / \frac{1}{2}(|I_{src}| + |I_{drn}|)$. Clipping: f_0 negatives set to zero (pre-clip min $\sim -3 \times 10^{-11}$, $\sim 46/328544$ nodes; moment change $< 10^{-4}$).

Software, commit, run commands. Python 3.13.15, NumPy 2.5.1, SciPy 1.18.0; code at the commit in Appendix B. From src/: `python coupled_she.py 60 50 -0.74` (recalibrated coupled solve); `python she2d_liang_overlay.py / she2d_liang_compare.py / she2d_paper_compare_figs.py` (Fig. 7 overlays, comparison plots); `python she2d_iv.py` (DD-proxy Fig. 8); `python she2d_iv_moment.py` (SHE velocity-moment I-V, Fig. 10); `python audit_result.py` (regression audit); `python coupled_converge.py 0.35 50` (damped-Picard baseline, superseded); `python coupled_anderson.py 80 66 6 0.5 28 1` (Anderson-accelerated coupled solve on a frozen H -grid, finest grid); `python she2d_anderson_fig.py` (Anderson vs Picard and the spatial sequence, Fig. 4).

Rev. 6 spatial-convergence study (§12). One parameterized self-similar driver imposes an identical grading law, one common frozen H -grid, and identical solver settings on every mesh, printing a run manifest (mesh law, measured dx/dy , common H -span, ε_{max} , all tolerances, git provenance) and hard-failing if a node is not sanctioned, if r_x/r_y deviate from $\sqrt{2}$, or if a converged ϕ leaves the common span. `solve_coupled` gained `phi_init/p_init` warm-start and `phi_span_H_override`; G_{tot} is tracked per outer iteration from the same G_{ii} field as $G_{ii,max}$ with nodal control-volume weights. Each invocation `python coupled_drain_richardson.py Nx Ny maxouter [warm.npz]` runs `maxouter` outer iterations and saves `she2d_richardson_NxxNy.npz` with the full diagnostic vector (last-4 $G_{ii,max}/G_{tot}/R_\phi$, dx/dy , H -hash, provenance).

Exact reproduction procedure (this is essential — a fixed budget does not converge these nodes). The archival nodes were produced by “continue unchanged until the plateau gate passes”: run a node, then *continue* it warm-started from its *own* saved npz under byte-identical settings until R_ϕ is at its floor and the last-4 relative spreads are $\lesssim 3 \times 10^{-4}$ in *both* G_{ii} and G_{tot} . Nodes are warm-start chained (100×70 from the converged 71×58 , 141×85 from the converged 100×70). This procedure is scripted end-to-end: `python run_richardson_trio.py` (which drives the three nodes in 60-iteration chunks, checking the gate after each and continuing as needed) followed by `python richardson_fit.py data/she2d_richardson_{71x58,100x70,141x85}.npz`; the whole pipeline from scratch is `run_from_scratch.bat`. For the record, the archival nodes reached the gate at $\approx 130/150/140$ total outer iterations for $71 \times 58/100 \times 70/141 \times 85$ respectively (the initial 60-iteration values were biased transients, shifting up to +18% on continuation — see the runtime note). `richardson_fit.py` gates on identical signatures + H -hash, computes the reversal ratio S_{rev} , and reports (or suppresses) the extrapolation. The three converged nodes and their logs are archived under `data/she2d_richardson*.npz` and `data/richardson*.log`.

Runtime and a plateau-gate note. On a single workstation the essential end-to-end rerun is ~ 26 – 27 h, dominated by the controlled trio (~ 23 h; the finest 141×85 node alone ~ 12 h at ~ 320 s/outer-iteration). The fine nodes run *sequentially* — at ~ 5 – 6 GB each on a 26 GB machine, three at once leaves too little headroom for the sparse work arrays — but they are independent, so across machines the critical path falls to the ~ 12 h finest node. A faster diagnostic rerun could terminate the coarser nodes at looser plateau quality; because the observed reversal is orders of magnitude larger than the plateau uncertainty ($S_{rev} \approx 874$), this is not expected to alter the qualitative verdict — though it is not verdict-neutral by construction. The archival result nevertheless uses the uniform $O(10^{-4})$ plateau gate on all three nodes.

B Provenance and audit history

Base code commit 0994290; Python 3.13.15, NumPy 2.5.1, SciPy 1.18.0. Key commits: f016e3c (locked one-way audit), 7aad998 (60×50 refinement), 65461ad (physics-split solver), 047da93 (coupled loop), 0994290 (recalibrated coupled 60×50). Four audit rounds fixed: over-claims, missing specs, junction geometry, N_c convergence, the reducible-collision zero-field- T_e bug (acoustic operator), the global H -grid, the contact-Maxwellian normalization (report typo; code correct), boundary/velocity/current extraction, and finally delivered the real impact-ionization operator, the coupled driver, and the physics-split solver. Revision 3 adds the quantitative comparison against Liang’s published figures (§11): the recalibrated coupled fields on Liang’s Fig. 7 contour levels, plus potential/hole (Figs. 5/6) and terminal I-V (Fig. 8) plots (`she2d_liang_compare.py`, `she2d_liang_overlay.py`, `she2d_liang_fig5.6.py`, `she2d_iv.py`), with one comparison table per plot (Tables 7–13). Revision 4 responds to an external audit: corrected the Fig. 7 contour levels (added 4800 K, removed a spurious $2.3 \times 10^{26} G_{ii}$ level) and regenerated the overlays; reframed the work as a *reduced* P1/SHE reproduction with a paper-to-code physics matrix (Table 2) and a calibration-vs-validation matrix (Table 3); added a one-way SHE velocity-moment I-V (Fig. 10); fixed the doping reference ([7]→[25]); and added the reproducibility package (Appendix A). The coupled-loop convergence study is now complete (§9, Fig. 3): 50 iterations reach 2.5 mV (not 2 mV) and reveal that G_{ii} does not converge in the outer loop, superseding the earlier snapshot values (overlays regenerated on the best-converged fields). the one-way SHE velocity-moment I-V campaign is complete (Fig. 10, Table 13). Revision 5 replaces the damped-Picard coupling with **Anderson acceleration** on a frozen H -grid (§10; `coupled_she.py accel="anderson"`, `driver_coupled_anderson.py`), which *converges* the self-consistent fixed point Picard could not. The converged $G_{ii,\max}$ over the 40×34/60×50/80×66 sequence is 8.1/2.2/1.5 × 10²⁶ (loop-converged but still mesh-dependent, ~15× below Liang); the overlays are regenerated on the converged 80×66 fields. The three-grid sequence is now a genuine spatial-convergence study; a converged spatial limit and explicit surface/impurity operators remained open at Rev. 5. Revision 6 carries out that spatial-convergence study (§12; `coupled_drain_richardson.py`, `richardson_fit.py`, `warm-start/phi_span_H_override` in `coupled_she.py`). Its outcome is decisive and negative: the uniform sequence was itself drain-under-resolved, and a controlled self-similar trio 71×58/100×70/141×85 (fixed law, common SHA1-identical H -grid, $\sqrt{2}$ refinement, all nodes plateau-gated to $O(10^{-4})$) is *significantly non-monotone* (2.990/2.135/2.413 × 10²⁵, $S_{\text{rev}} \approx 874$), so no continuum G_{ii} can be inferred from this trio and spatial convergence is not demonstrated. A prominent secondary finding is that the earlier 60-iteration node values were materially biased fixed-point transients (up to +18%), motivating an observable-plateau validity gate. *Threshold provenance*: an interim Rev. 5.1 draft stated the observable-plateau gate as “sub-10⁻⁴”; on review it was standardized to the final operational form $O(10^{-4})$, $\lesssim 3 \times 10^{-4}$, applied identically to all three nodes and to both G_{ii} and G_{tot} (this admits G_2 ’s $2.9 \times 10^{-4} G_{ii}$ spread). The verdict is insensitive to this distinction: $S_{\text{rev}} \approx 874$ exceeds any plateau-band choice in this range by orders of magnitude. This work was prepared on the local `claude/init-e9d010` working tree based on 676e74f; it has no external audit round yet.

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